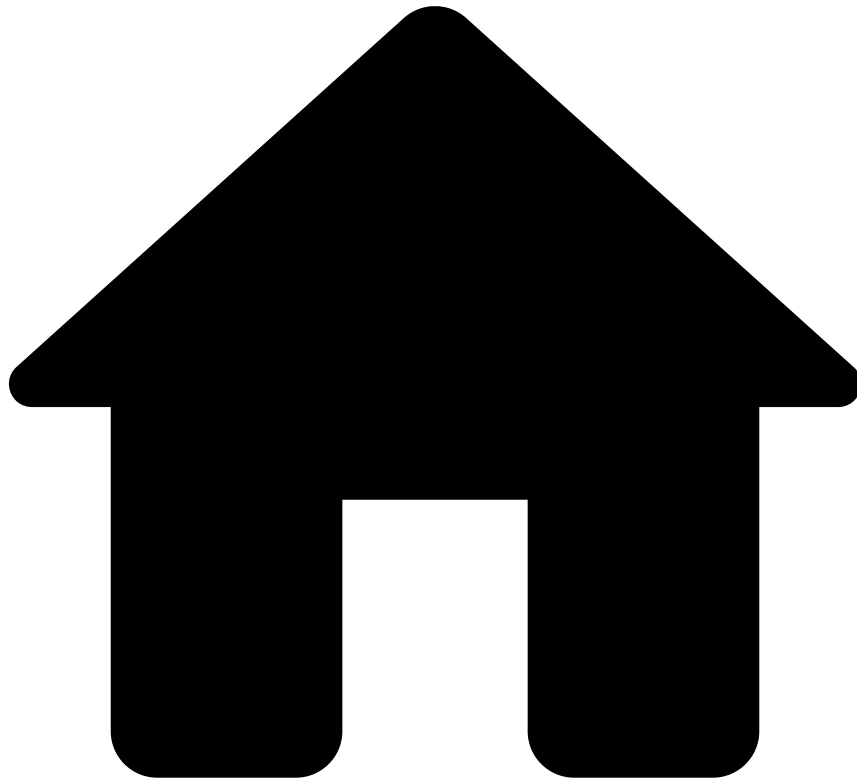


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- [Creating a Data Science project](#)
- [Processing your data](#)
- [Analyzing and sampling your data](#)
- Creating a Stratified Sample

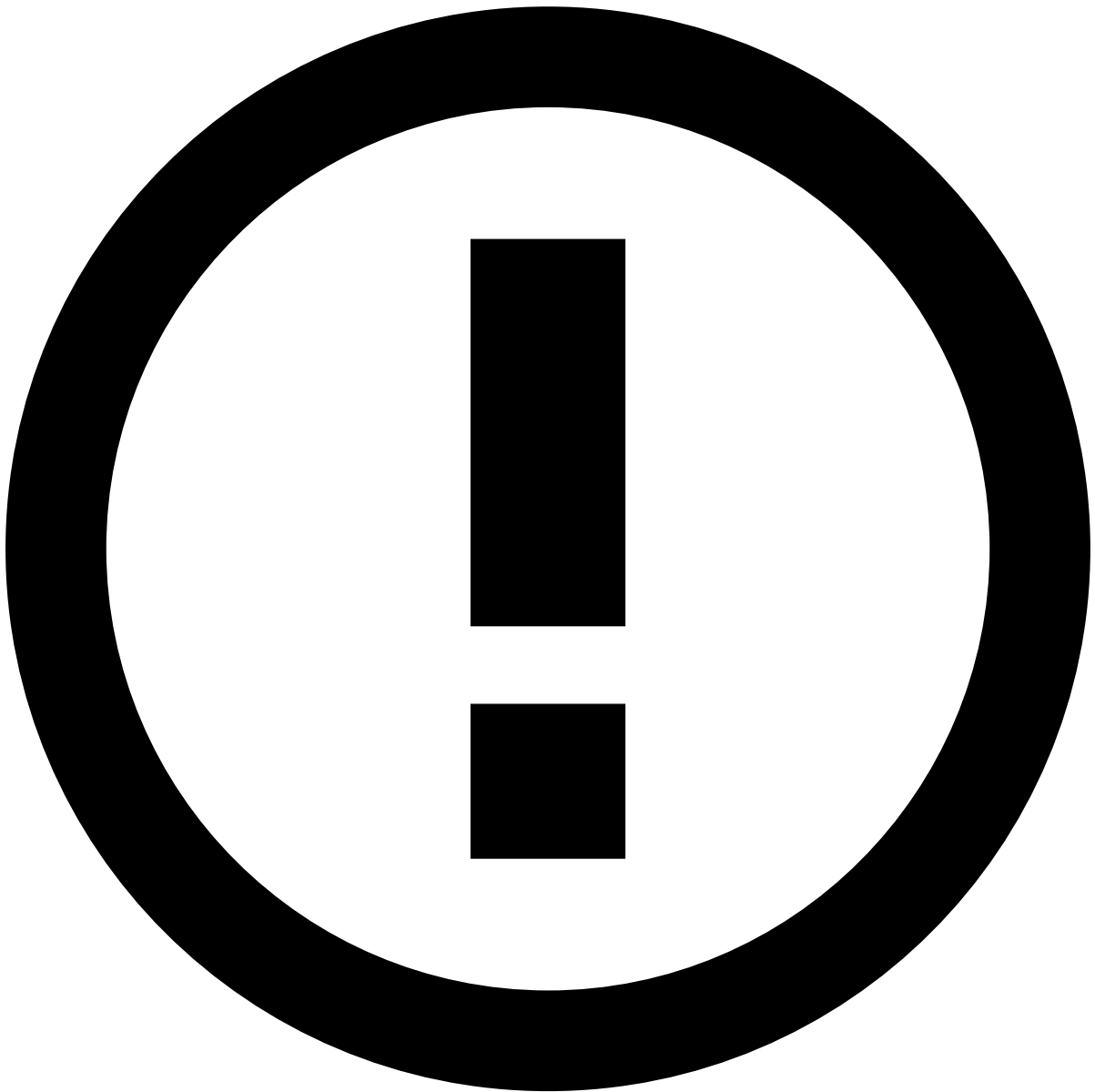
On this page

Creating a Stratified Sample

Was this page helpful? 🗨️

Generating a stratified sample is very common in fraud detection scenarios and represents the basis for [Feature Engineering](#).

Pulse enables you to design a [Plan](#) and generate a stratified sample from terabytes of historical data, with little need for manual actions, technical proficiency in external tools and access to cluster infrastructure.



info

Alternatively, you can perform these actions from the JupyterLab user interface. Standard notebooks are available within [JupyterLab](#) – in particular the Sampling Notebook, which demonstrates how to apply efficient sampling strategies over large datasets using Feedzai's sampling library.

How to

The following instructions exemplify a way to generate a stratified sample for a card fraud scenario. This plan extracts some cards that never had fraud, and all fraudulent cards, out of the original data, and obtains all transactions for those cards.

To design a plan and generate a stratified sample for the *Transactions* data source, execute the following actions:

1. In the **Data Science tab**, **Add** a new plan.
2. Set the plan as [Non-Deployable](#).

3. Design your Stratified Sample plan as displayed in the following image:

Configure the elements as follows:

1. Edit the **Source** element and specify the date source. For example, *Cleaned Transactions*.
2. Edit the **Map** element to add a new field to the date source, as follows:

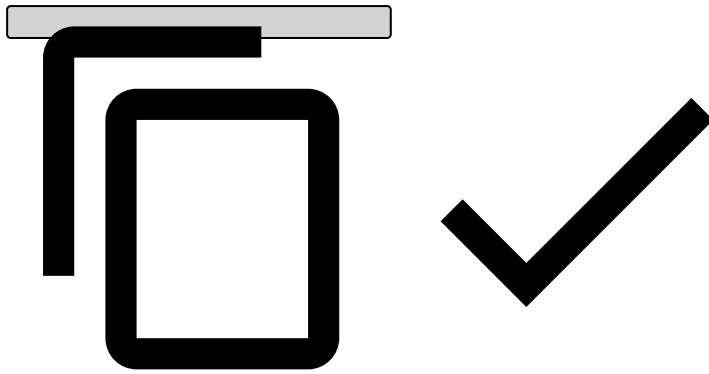


Clipboard

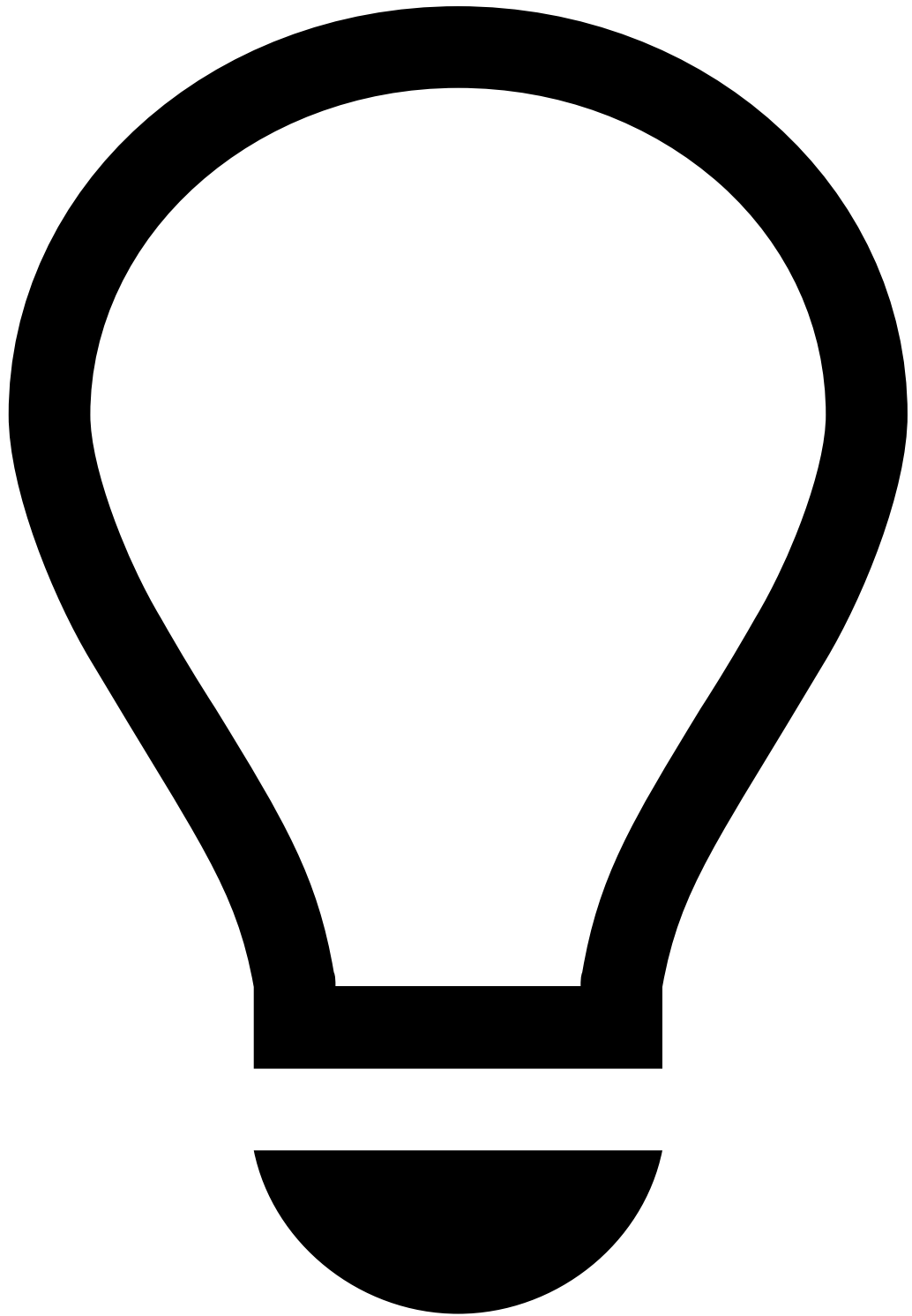
You can also copy Operators to the clipboard in order to paste them in other Plans.

1. Select the **Fields to Add** tab and name the new field as *isFraud*.
2. Write the following expression for the field value:

```
if (Fraud == 'True') then 1 else 0
```



3. Add the field to the list of fields to be added to the data source.
3. Edit Edit the **Group By** element to specify grouping by *AccountID* (represents the card number in this case) and calculate the *Sum* aggregation of the *isFraud* field.

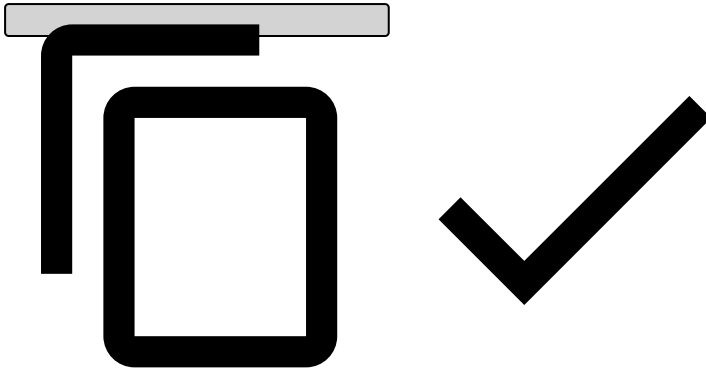


Checkpoint

At this stage you have the number of fraud occurrences for each card number, calculated into the *sumFraud* field. The plan workflow will branch to segment fraudulent cards from cards that never observed fraud.

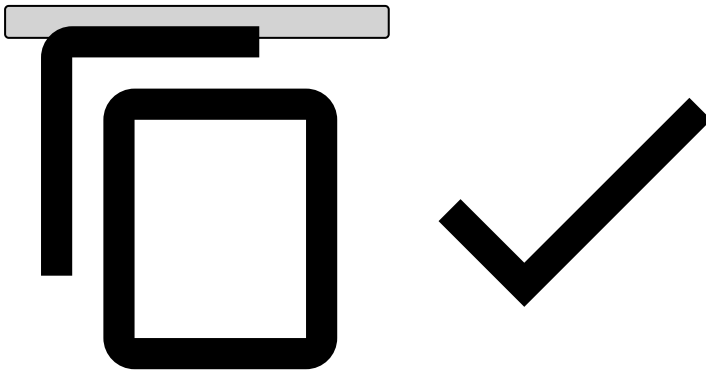
4. Edit Edit the **Filter** element with the following expression to obtain only the card numbers **with** fraudulent transactions.

```
sumFraud > 0
```

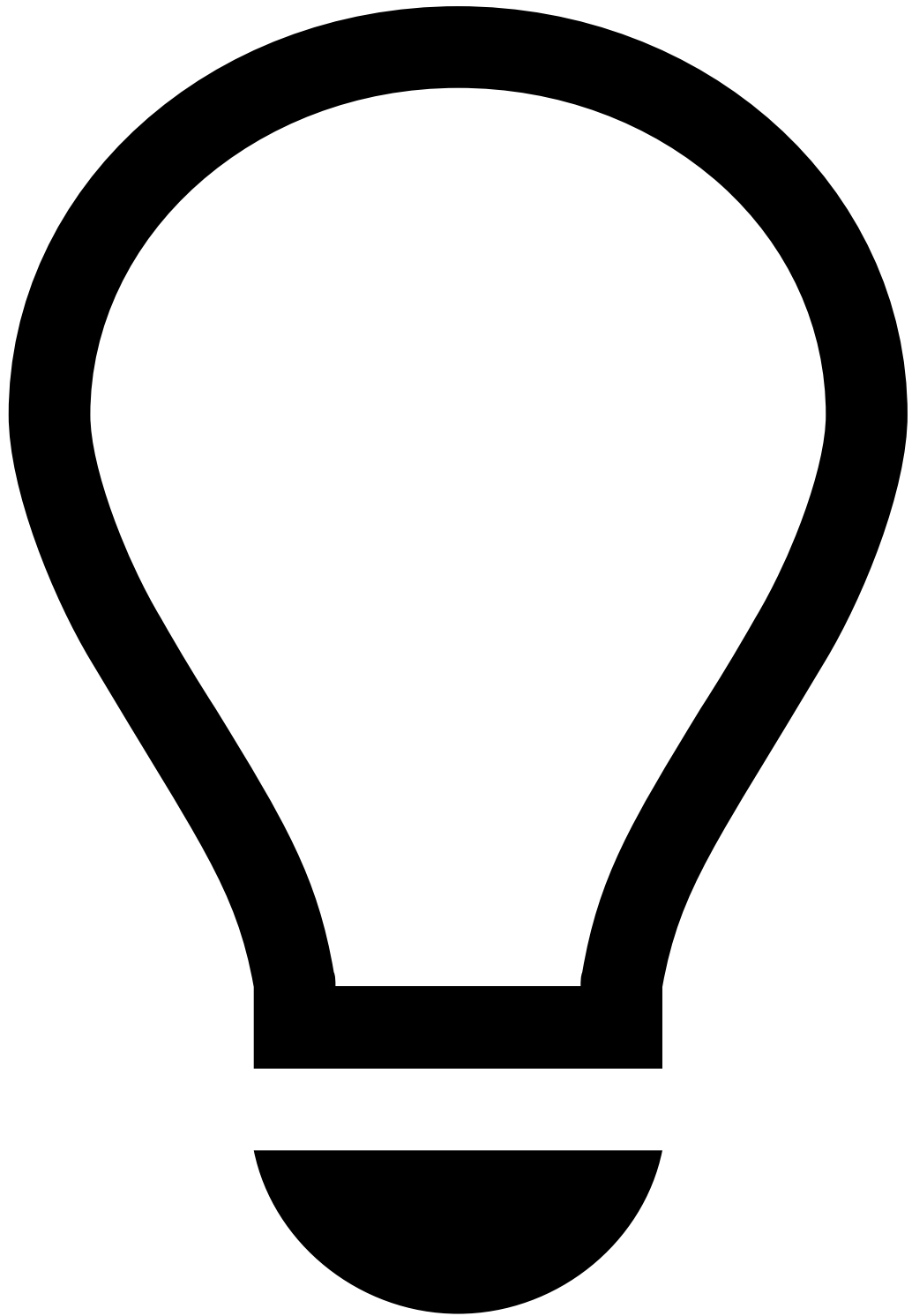


5. Edit Edit the **Filter** element with the following expression to obtain only the card numbers **without** fraudulent transactions.

```
sumFraud == 0
```



6. Edit Edit the **Sample** element to extract only 5% of all cards **without** fraudulent transactions.



Checkpoint

At this stage you have all fraudulent card numbers separate from a smaller subset of all cards that never had fraud. The plan workflow will recover the complete transactions for all cards in both branches and re-merge them together as a single dataset.

7. Edit Edit the **Join** element to perform a join operation, by *AccountID*, with the original data and recover the original transactions for those cards. You now have all transactions for cards where fraud **occurred**.
8. Edit Edit the **Join** element to perform a join operation, by *AccountID*, with the original data and recover the original transactions for those cards. You now have all transactions for cards where fraud **never occurred**.
9. Execute a **Union** of both **Join** elements.
10. Save the results of your plan to a data **Sink**. For example, a new data source named *Stratified Sample*.

Finally:

1. Run the plan to execute all actions on the original data source and generate an improved one.
 1. Define the **Data Source with ID source** as the original data source. For example, *Cleaned Transactions*.
 2. Expand the Advanced Options to have access to additional configurations:
 1. [Job Profile](#): Allows you to choose a profile containing properties that best fit the **Spark Job** that you want to run. Check the page [Selecting Profiles](#) to learn more about this functionality.
 2. [Data Partitioning](#): If you have [Metrics Operators](#) in your plan, this functionality allows to choose the **Data Partitioning** option that enables **Spark** to re-organize the data into different partitions as homogenous and well distributed as possible. Pulse allows you to choose the options: **Data-Aware Optimizer**, **Hash-Based**, and **Manual Partition**. These options are described in detail in the [Data Partitioning](#) page.



Metrics Operators

The **Data Partitioning** option takes effect only if you have metrics operators in your plan. Otherwise, this option does not affect the performance of the plan execution.

3. Configure the **Data Sink with ID sink** as the new and clean data source. For example, *Stratified Sample*.
 4. **Run** the plan to generate a new data source, in Pulse, named *Stratified Sample*. Pulse also saves an output file with the transformed data to your filesystem.
2. [Analyze](#) the *Stratified Sample* data source to check the distribution of fraud and non-fraud transactions.

Next Steps

When you have sampled and understood your data, you are ready to move on to [designing features](#) for your model.